# Nmap Post Port Scans

## Service Detection

`-sV` to your Nmap command will collect and determine service and version information for the open ports

control the intensity with `--version-intensity LEVEL` where the level ranges between 0, the lightest, and 9, the most complete. `-sV --version-light` has an intensity of 2, while `-sV --version-all` has an intensity of 9.

```
Pentester Terminal

pentester@TryHackMe$ sudo nmap -sV 10.130.137.198

Starting Nmap 7.60 ( https://nmap.org ) at 2021-09-10 05:03 BST
Nmap scan report for 10.130.137.198
Host is up (0.0040s latency).
Not shown: 995 closed ports
PORT    STATE SERVICE VERSION
22/tcp  open  ssh     OpenSSH 6.7p1 Debian 5+deb8u8 (protocol 2.0)
25/tcp  open  smtp    Postfix smtpd
80/tcp  open  http    nginx 1.6.2
110/tcp open  pop3    Dovecot pop3d
111/tcp open  rpcbind 2-4 (RPC #100000)
MAC Address: 02:A0:E7:B5:B6:C5 (Unknown)
Service Info: Host:  debra2.thm.local; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.or
Nmap done: 1 IP address (1 host up) scanned in 8.40 seconds
```

# OS Detection and Traceroute

## OS Detection

Nmap can detect the Operating System (OS) based on its behaviour and any telltale signs in its responses. OS detection can be enabled using `-O`

*uppercase O* as in OS. In this example, we ran `nmap -sS -O MACHINE_IP`

```
pentester@TryHackMe$ sudo nmap -sS -O MACHINE_IP

Starting Nmap 7.60 ( https://nmap.org ) at 2021-09-10 05:04 BST
Nmap scan report for MACHINE_IP
Host is up (0.00099s latency).
Not shown: 994 closed ports
PORT    STATE SERVICE
22/tcp  open  ssh
25/tcp  open  smtp
80/tcp  open  http
110/tcp open  pop3
111/tcp open  rpcbind
143/tcp open  imap
MAC Address: 02:A0:E7:B5:B6:C5 (Unknown)
Device type: general purpose
Running: Linux 3.X
OS CPE: cpe:/o:linux:linux_kernel:3.13
OS details: Linux 3.13
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/sub
Nmap done: 1 IP address (1 host up) scanned in 3.91 seconds
```

## Trace Route

If you want Nmap to find the routers between you and the target, just add `--traceroute`

Nmap’s traceroute starts with a packet of high TTL and keeps decreasing it.

`nmap -sS --traceroute MACHINE_IP`

```
pentester@TryHackMe$ sudo nmap -sS --traceroute MACHINE_IP

Starting Nmap 7.60 ( https://nmap.org ) at 2021-09-10 05:05 BST
Nmap scan report for MACHINE_IP
Host is up (0.0015s latency).
Not shown: 994 closed ports
PORT    STATE SERVICE
22/tcp  open  ssh
25/tcp  open  smtp
80/tcp  open  http
110/tcp open  pop3
111/tcp open  rpcbind
143/tcp open  imap
MAC Address: 02:A0:E7:B5:B6:C5 (Unknown)

TRACEROUTE
HOP RTT     ADDRESS
1   1.48 ms MACHINE_IP

Nmap done: 1 IP address (1 host up) scanned in 1.59 seconds
```

## Nmap Scripting Engine (NSE)

A part of Nmap, Nmap Scripting Engine (NSE) is a Lua interpreter that allows Nmap to execute Nmap scripts written in Lua language. However, we don’t need to learn Lua to make use of Nmap scripts.

Pentester Terminal

```
pentester@AttackBox /usr/share/nmap/scripts# ls http*
http-adobe-coldfusion-apsa1301.nse      http-passwd.nse
http-affiliate-id.nse                   http-php-version.nse
http-apache-negotiation.nse             http-phpmyadmin-dir-traversal.nse
http-apache-server-status.nse           http-phpself-xss.nse
http-aspnet-debug.nse                   http-proxy-brute.nse
http-auth-finder.nse                    http-put.nse
http-auth.nse                           http-qnap-nas-info.nse
http-avaya-ipoffice-users.nse           http-referer-checker.nse
http-awstatstotals-exec.nse             http-rfi-spider.nse
http-axis2-dir-traversal.nse            http-robots.txt.nse
http-backup-finder.nse                  http-robtex-reverse-ip.nse
http-barracuda-dir-traversal.nse        http-robtex-shared-ns.nse
http-brute.nse                          http-security-headers.nse
http-cakephp-version.nse                http-server-header.nse
http-chrono.nse                         http-shellshock.nse
http-cisco-anyconnect.nse               http-sitemap-generator.nse
http-coldfusion-subzero.nse             http-slowloris-check.nse
http-comments-displayer.nse             http-slowloris.nse
http-config-backup.nse                  http-sql-injection.nse
http-cookie-flags.nse                   http-stored-xss.nse
http-cors.nse                           http-svn-enum.nse
```

You can choose to run the scripts in the default category using `--script=default` or simply adding `-sC`. In addition to [default(opens in new tab)](), categories include auth, broadcast, brute, default, discovery, dos, exploit, external, fuzzer, intrusive, malware, safe, version, and vuln.

| Script Category | Description |
|---|---|
| `auth` | Authentication related scripts |
| `broadcast` | Discover hosts by sending broadcast messages |
| `brute` | Performs brute-force password auditing against logins |
| `default` | Default scripts, same as `-sC` |
| `discovery` | Retrieve accessible information, such as database tables and DNS names |
| `dos` | Detects servers vulnerable to Denial of Service (DoS) |
| `exploit` | Attempts to exploit various vulnerable services |
| `external` | Checks using a third-party service, such as Geoplugin and Virustotal |
| `fuzzer` | Launch fuzzing attacks |
| `intrusive` | Intrusive scripts such as brute-force attacks and exploitation |
| `malware` | Scans for backdoors |
| `safe` | Safe scripts that won't crash the target |
| `version` | Retrieve service versions |
| `vuln` | Checks for vulnerabilities or exploit vulnerable services |

`sudo nmap -sS -sC`

`-sC` will ensure that Nmap will execute the default scripts following the SYN scan.

Pentester Terminal

```
pentester@TryHackMe$ sudo nmap -sS -sC 10.128.171.61

Starting Nmap 7.60 ( https://nmap.org ) at 2021-09-10 05:08 BST
Nmap scan report for ip-10-10-161-170.eu-west-1.compute.internal (10.10.161.170)
Host is up (0.0011s latency).
Not shown: 994 closed ports
PORT    STATE SERVICE
22/tcp  open  ssh
| ssh-hostkey:
|   1024 d5:80:97:a3:a8:3b:57:78:2f:0a:78:ae:ad:34:24:f4 (DSA)
|   2048 aa:66:7a:45:eb:d1:8c:00:e3:12:31:d8:76:8e:ed:3a (RSA)
|   256 3d:82:72:a3:07:49:2e:cb:d9:87:db:08:c6:90:56:65 (ECDSA)
|_  256 dc:f0:0c:89:70:87:65:ba:52:b1:e9:59:f7:5d:d2:6a (EdDSA)
25/tcp  open  smtp
|_smtp-commands: debra2.thm.local, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS,
| ssl-cert: Subject: commonName=debra2.thm.local
| Not valid before: 2021-08-10T12:10:58
|_Not valid after:  2031-08-08T12:10:58
|_ssl-date: TLS randomness does not represent time
80/tcp  open  http
|_http-title: Welcome to nginx on Debian!
110/tcp open  pop3
|_pop3-capabilities: RESP-CODES CAPA TOP SASL UIDL PIPELINING AUTH-RESP-CODE
111/tcp open  rpcbind
| rpcinfo:
|   program version   port/proto  service
|   100000  2,3,4        111/tcp  rpcbind
|   100000  2,3,4        111/udp  rpcbind
|   100024  1          38099/tcp  status
|_  100024  1          54067/udp  status
143/tcp open  imap
|_imap-capabilities: LITERAL+ capabilities IMAP4rev1 OK Pre-login ENABLE have LOGIN
MAC Address: 02:A0:E7:B5:B6:C5 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 2.21 seconds
```

specify the script by name using `--script "SCRIPT-NAME"` or a pattern such as `--script "ftp*"`, which would include `ftp-brute`

```
sudo nmap -sS -n --script "http-date" 10.128.171.61
```

```
root@ip-10-128-105-20:~# sudo nmap -sS -n --script "http-date" 10.128.171.61
sudo: unable to resolve host ip-10-128-105-20: Name or service not known
Starting Nmap 7.80 ( https://nmap.org ) at 2026-04-26 13:59 BST
Nmap scan report for 10.128.171.61
Host is up (0.0052s latency).
Not shown: 991 closed ports
PORT    STATE SERVICE
22/tcp  open  ssh
25/tcp  open  smtp
53/tcp  open  domain
80/tcp  open  http
|_http-date: Sun, 26 Apr 2026 12:59:13 GMT; -1s from local time.
110/tcp open  pop3
111/tcp open  rpcbind
143/tcp open  imap
993/tcp open  imaps
995/tcp open  pop3s

Nmap done: 1 IP address (1 host up) scanned in 1.08 seconds
```

## Saving the Output

The number of files can quickly grow and hinder your ability to find a previous scan result. The three main formats are:

1. Normal
2. Grepable (`grep` able)
3. XML

### Normal

You can save your scan in normal format by using `-oN FILENAME`; N stands for normal

```
Pentester Terminal

pentester@TryHackMe$ cat MACHINE_IP_scan.nmap
# Nmap 7.60 scan initiated Fri Sep 10 05:14:19 2021 as: nmap -sS -sV -O -oN MACHINI
Nmap scan report for 10.128.130.35
Host is up (0.00086s latency).
Not shown: 994 closed ports
PORT    STATE SERVICE VERSION
22/tcp  open  ssh     OpenSSH 6.7p1 Debian 5+deb8u8 (protocol 2.0)
25/tcp  open  smtp    Postfix smtpd
80/tcp  open  http    nginx 1.6.2
110/tcp open  pop3    Dovecot pop3d
111/tcp open  rpcbind 2-4 (RPC #100000)
143/tcp open  imap    Dovecot imapd
MAC Address: 02:A0:E7:B5:B6:C5 (Unknown)
Device type: general purpose
Running: Linux 3.X
OS CPE: cpe:/o:linux:linux_kernel:3.13
OS details: Linux 3.13
Network Distance: 1 hop
Service Info: Host:  debra2.thm.local; OS: Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https:/
# Nmap done at Fri Sep 10 05:14:28 2021 -- 1 IP address (1 host up) scanned in 9.9
```

## Grepable

can save the scan result in grepable format using `-oG FILENAME`

```
Pentester Terminal

pentester@TryHackMe$ cat MACHINE_IP_scan.gnmap
# Nmap 7.60 scan initiated Fri Sep 10 05:14:19 2021 as: nmap -sS -sV -O -oG MACHINE
Host: 10.128.130.35 Status: Up
Host: 10.128.130.35 Ports: 22/open/tcp//ssh//OpenSSH 6.7p1 Debian 5+deb8u8 (protoco
# Nmap done at Fri Sep 10 05:14:28 2021 -- 1 IP address (1 host up) scanned in 9.9
```

`grep KEYWORD TEXT_FILE` ; this command will display all the lines containing the provided keyword

```
pentester@TryHackMe$ grep http MACHINE_IP_scan.nmap
80/tcp   open   http     nginx 1.6.2
OS and Service detection performed. Please report any incorrect results at https://
```

```
pentester@TryHackMe$ grep http MACHINE_IP_scan.gnmap
Host: 10.128.130.35 Ports: 22/open/tcp//ssh//OpenSSH 6.7p1 Debian 5+deb8u8 (protoco
```

## XML

You can save the scan results in XML format using `-oX FILENAME`. The XML format would be most convenient to process the output in other programs. Conveniently enough, you can save the scan output in all three formats using `-oA FILENAME` to combine `-oN`, `-oG`, and `-oX` for normal, grepable, and XML.

## Script Kiddie

useless if you want to search the output for any interesting keywords or keep the results for future reference. However, you can use it to save the output of the scan `nmap -sS 127.0.0.1 -oS FILENAME`, display the output filename, and look 31337 in front of friends who are not tech-savvy.

Pentester Terminal

```
pentester@TryHackMe$ cat MACHINE_IP_scan.kiddie

$tart!ng nMaP 7.60 ( httpz://nMap.0rG ) at 2021-09-10 05:17 B$T
Nmap scan rEp0rt f0r |p-10-10-161-170.EU-w3$t-1.C0mputE.intErnaL (10.10.161.170)
HOSt !s uP (0.00095s LatEncy).
N0T $H0wn: 994 closed pOrtS
PoRT     st4Te SeRViC3 VERS1on
22/tcp   Open  ssH     Op3n$$H 6.7p1 Deb|an 5+dEb8u8 (pr0t0COl 2.0)
25/tCp   Op3n  SmTp    P0$Tf!x Smtpd
80/tcp   0p3n  http    Ng1nx 1.6.2
110/tCP  0pen  pOP3    d0v3coT P0p3D
111/TcP  op3n  RpcbInd 2-4 (RPC #100000)
143/Tcp  opEn  Imap    Dovecot 1mApd
mAC 4Ddr3sz: 02:40:e7:B5:B6:c5 (Unknown)
Netw0rk d!stanc3: 1 h0p
$3rv1c3 InFO: Ho$t:  dEBra2.thM.lOcal; 0s: Linux; cPe: cP3:/0:linux:l|nux_k3rnel

0S and servIc3 D3tEcti0n pErf0rm3d. Plea$e r3p0rt any !nc0RrecT rE$ultz at hTtpz:/
Nmap d0nE: 1 |P addr3SS (1 hoSt up) $CaNnEd !n 21.80 s3c0Ndz
```